

Eqs. (9.43)–(9.48)). However, for the general case, the temperature coefficient of the resistor-MOSFET voltage divider is determined using

$$TCV_{REF} = \frac{1}{V_{REF}} \cdot \frac{\partial V_{REF}}{\partial T} \quad (23.3)$$

and if we assume $VDD \gg V_{REF}$, then

$$TCV_{REF} = \frac{1}{V_{REF}} \left[V_{THN} \cdot TCV_{THN} - \frac{1}{2} \sqrt{\frac{2L_1}{W_1} \cdot \frac{VDD}{R \cdot KP(T)}} \cdot \left[\frac{1}{R} \frac{\partial R}{\partial T} - \frac{1.5}{T} \right] \right] \quad (23.4)$$

Example 23.1

Estimate the temperature performance of the resistor-MOSFET voltage references seen in Fig. 23.3. Assume that the temperature coefficient of the resistor is 2,000 ppm/C, the long-channel process is used where $KP_n = 120 \mu\text{A}/\text{V}^2$, and the nominal VDD is 5 V. Use simulations to verify the answers. Also show how the reference voltages change with VDD .



Figure 23.3 Resistor-MOSFET references used in Ex. 23.1

For the reference made using the 1MEG resistor, Fig. 23.3a, the current that flows in the circuit is

$$I = \frac{VDD - V_{REF}}{10^6} = \frac{KP_n}{2} \cdot \frac{W}{L} (V_{REF} - V_{THN})^2$$

which can be solved to determine I is around 4 μA and

$$V_{REF} = V_{GS} \approx 900 \text{ mV}$$

From Sec. 9.1.3 the rate the threshold voltage changes with temperature is

$$\frac{\partial V_{THN}}{\partial T} \approx -1 \text{ mV}/^\circ\text{C} \approx \frac{\partial V_{REF}}{\partial T}$$

The temperature coefficient of the reference voltage is

$$TCV_{REF} = \frac{1}{V_{REF}} \frac{\partial V_{REF}}{\partial T} = \frac{-0.001}{0.9} = -1,111 \text{ ppm}/^\circ\text{C} \quad (23.5)$$

and thus

$$V_{REF}(T) = V_{REF} \cdot (1 + TCV_{REF}(T - T_0)) = 0.9 \cdot (1 - 0.00111(T - 25)) \quad (23.6)$$

where we assume that the threshold voltage was measured at 25 °C. The simulation results are seen in Fig. 23.4. Note that at higher temperatures the threshold voltage decreases and thus so does V_{REF} . Further note that a change in temperature of 25 °C corresponds to a change in the reference voltage of 31.25 mV ($= 25 \cdot 0.00125$).

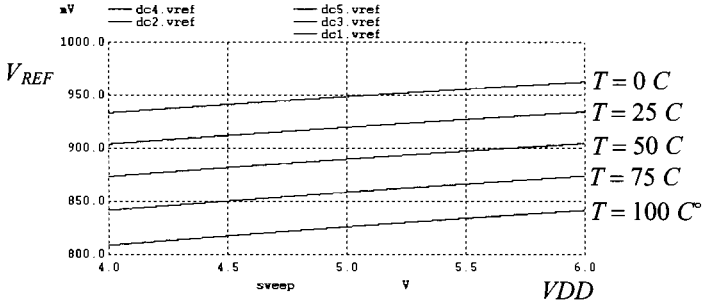


Figure 23.4 How the reference voltage changes with VDD and temperature for the reference in Fig. 23.3a.

To estimate the temperature behavior of the circuit in Fig. 23.3b, we must first determine the value of V_{REF} . Using Eq. (23.2), we can write

$$V_{REF} = V_{THN} + \sqrt{\frac{2(VDD - V_{REF})}{R \cdot \beta_1}} = 0.8 + \sqrt{\frac{2(5 - V_{REF})}{10k \cdot \frac{10}{2} \cdot 120\mu A/V^2}}$$

After a few iterations, we can determine $V_{REF} \approx 1.85 \text{ V}$. The temperature coefficient is calculated using Eq. (23.4)

$$TCV_{REF} = \frac{1}{1.85} \cdot \left[-1 \text{ mV/}^\circ\text{C} - \frac{1}{2} \sqrt{\frac{2 \cdot 2}{10} \cdot \frac{5}{10k \cdot 120 \mu A/V^2}} \cdot \left[0.002 - \frac{1.5}{300} \right] \right]$$

which evaluates to $TCV_{REF} = 500 \text{ ppm/}^\circ\text{C}$. The simulation results are seen in Fig. 23.5. Note how, when comparing Figs. 23.4 and 23.5, one reference circuit performs well with regard to temperature (23.3b), while the other reference circuit (23.3a) performs well with regard to VDD variations. ■

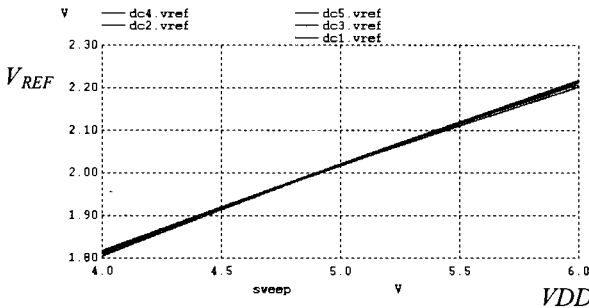


Figure 23.5 How the reference voltage changes with VDD and temperature for the reference in Fig. 23.3b.

A modification to the basic resistor-MOSFET divider is shown in Fig. 23.6. The reference voltage in this circuit is given by

$$V_{REF} = V_{GS} \left(\frac{R_1}{R_2} + 1 \right) \quad (23.7)$$

noting the ratio of the resistors (and so their temperature behavior or sheet resistance shift with process variations) doesn't affect the reference voltage. When V_{GS} is designed to be approximately V_{THN} , the resulting reference circuit is said to be a threshold-voltage multiplier reference circuit.

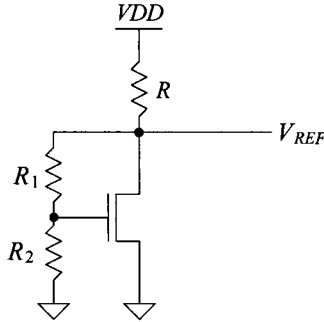


Figure 23.6 Modification of the resistor-MOSFET voltage divider.

23.1.2 The MOSFET-Only Voltage Divider

The MOSFET-only voltage divider shown in Fig. 23.2 generates a reference voltage that is equal to the voltage on the gates of the MOSFETs with respect to ground. Since $I_{D1} = I_{D2}$, we can write

$$\frac{\beta_1}{2}(V_{REF} - V_{THN})^2 = \frac{\beta_2}{2}(VDD - V_{REF} - V_{THP})^2 \quad (23.8)$$

or the reference voltage is given by

$$V_{REF} = \frac{VDD - V_{THP} + \sqrt{\frac{\beta_1}{\beta_2}} \cdot V_{THN}}{\sqrt{\frac{\beta_1}{\beta_2}} + 1} \quad (23.9)$$

or knowing the desired reference voltage and the power supply voltage

$$\frac{\beta_1}{\beta_2} = \left[\frac{VDD - V_{REF} - V_{THP}}{V_{REF} - V_{THN}} \right]^2 \quad (23.10)$$

The temperature dependence of the MOSFET-only voltage divider, assuming the temperature dependence of the ratio of the transconductance parameters, $\frac{\beta_1}{\beta_2}$, is negligible, is given by

$$TCV_{REF} = \frac{1}{V_{REF}} \cdot \frac{\partial V_{REF}}{\partial T} = \frac{1}{V_{REF}} \cdot \frac{1}{\sqrt{\frac{\beta_1}{\beta_2}} + 1} \cdot \left[\frac{\partial(-V_{THP})}{\partial T} + \sqrt{\frac{\beta_1}{\beta_2}} \frac{\partial V_{THN}}{\partial T} \right] \quad (23.11)$$

amount of signal fed back and added to the input. Because M3 and M4 are *inverting* common-source amplifiers, we want the signal fed back to the + input of the amplifier to be larger than the signal fed back to the – input of the amplifier. Since the (AC) currents flowing in M3 and M4 are equal and the small-signal resistance of $M2 + R$ is larger than the small-signal resistance of M1, we connect M2 to the + amplifier input.

Forcing the Same Current through Each Side of the Reference

The simplest method to force the same current through each side of a reference is to use a simple current mirror, Fig. 23.8a. As we saw in Fig. 20.15, the low-output resistance found in short-channel devices makes the resulting reference very sensitive to changes in V_{DD} . By cascoding the current mirror, Fig. 23.8b, the currents can be made more equal and the sensitivity to V_{DD} can be reduced. The problem with using a cascode current mirror is that the minimum allowable V_{DD} increases. Consider the following.



Figure 23.8 Using a current mirror to force the same current through each side of a reference.

Example 23.2

Resimulate the BMR in Fig. 20.18 using a cascode current mirror for the PMOS devices as seen in Fig. 23.9.

Using the cascode current mirror, we expect the currents through each branch of the reference to be very nearly equal. As seen in the simulation results in Fig. 23.10, the currents *are* equal. However, they are not constant and independent of changes in V_{DD} . Similarly, if we were to use V_{biasn} as a reference voltage (see Sec. 20.1.5), then V_{biasn} would show a large sensitivity to changes in V_{DD} . The problem with cascoding only the PMOS devices is that while the currents through each branch are equal (but not constant), the variation with V_{DD} is still present. The voltages at the drains of M1 and M2 change with V_{DD} . To reduce this voltage variation, we might consider cascoding the NMOS devices as well (see Fig. 23.11). The PMOS devices are still used to force the same current through each side of the reference, while now the NMOS cascode stack is used to keep the voltages across M1 and M2 constant with changes in V_{DD} . As the simulation results, Fig. 23.12, show, the currents do stabilize but at a voltage 20% higher than V_{DD} . Clearly, cascoding devices won't be useful in a short-channel CMOS process (and so we'll stick with using the added amplifier to both set the currents

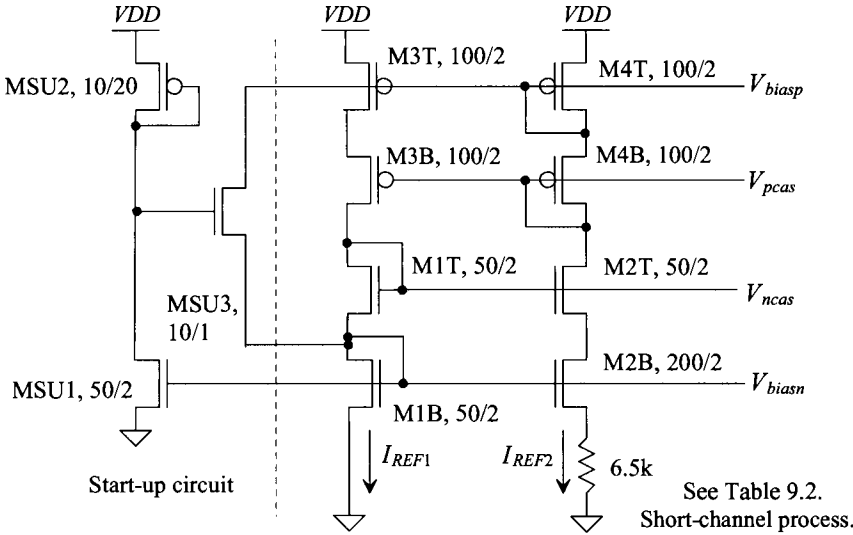


Figure 23.11 Cascoding both NMOS and PMOS devices in the BMR circuit.

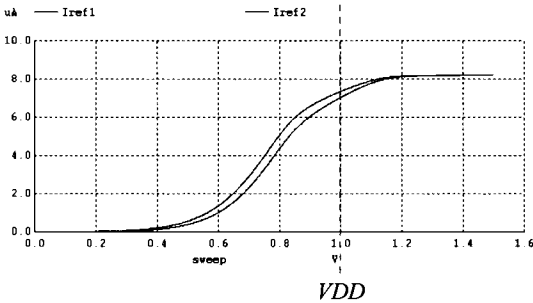


Figure 23.12 Simulating the behavior of the currents in the BMR of Fig. 23.11.

current flows in the reference. It's important to ensure that the start-up circuit doesn't affect normal operation or draw too much current from VDD . For the start-up circuit seen in Figs. 23.9 and 23.11, we use the V_{biasn} to set the current drawn by the bias circuit. In normal operation, MSU3 should be off (it should have a negative V_{GS}). ■

Example 23.3

Using the topology seen in Fig. 20.22, design a voltage reference with, ideally, zero temperature coefficient. Simulate the design to determine the reference's sensitivity to changes in temperature and VDD . Comment on the repeatability of the V_{REF} (with process variations from one process run to the next) and methods (and concerns) with trimming V_{REF} . Assume $TCR = 0.002$ (2,000 ppm/C).

We begin by writing the long-channel equation, Eq. (20.40), for the resistance needed for zero temperature coefficient (ZTC)

$$R = \frac{2}{\frac{\partial V_{THN}}{\partial T} \cdot KP_n \cdot \frac{W}{L}} \left(1 - \frac{1}{\sqrt{K}} \right) \cdot \left(\frac{1}{R} \frac{\partial R}{\partial T} + \frac{1}{KP_n} \cdot \frac{\partial KP_n}{\partial T} \right) \quad (23.16)$$

While this equation won't directly apply for the design in a short-channel process, we can use it, with simulations, to help point us to the factors that influence the temperature behavior of the reference. Substituting in the appropriate numbers with $K = 4$, we get

$$R = \frac{1}{(-0.0006) \cdot KP_n \cdot \frac{W}{L}} \cdot \left((0.002) - \frac{1.5}{300} \right) \quad (23.17)$$

or

$$R = \frac{5}{KP_n \cdot \frac{W}{L}} \quad (23.18)$$

To move the reference towards a ZTC, let's set the resistor to nominally 5.5k (the same value we used before, see Fig. 20.22 and the associated discussion) and adjust the widths and lengths of the NMOS devices until the simulations show good temperature behavior (keeping in mind that we must keep the W/L of M2 K times the W/L of M1). The resulting circuit is seen in Fig. 23.13. Simulation results are seen in Fig. 23.14. (Note that the currents and gate-source voltages have nothing to do with the values listed in Table 9.2.) The reference voltage is nominally 500 mV ($= VDD/2$). The reference, according to simulations, moves 40 mV over a 100 °C temperature change ($-400 \mu\text{V}/^\circ\text{C}$ or a TC of $-800 \text{ ppm}/^\circ\text{C}$, not that great when compared to the references described later). Further, the reference is fairly insensitive to variations in VDD once VDD gets above 600 mV.

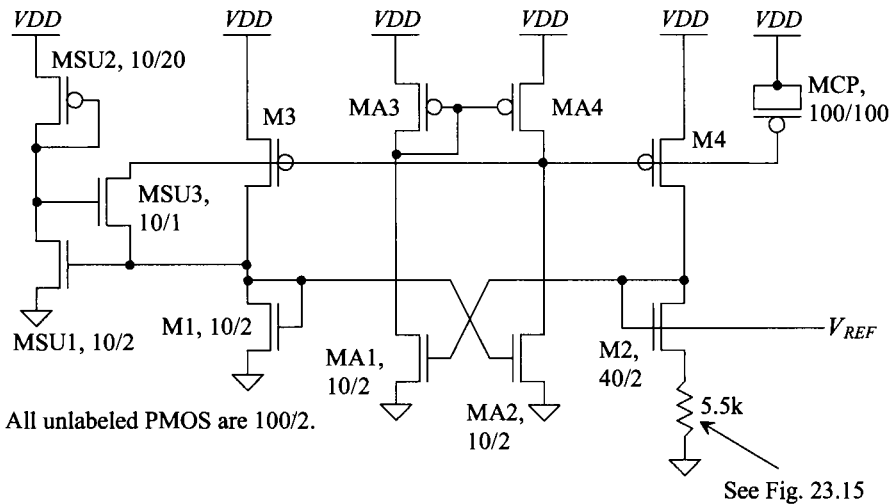


Figure 23.13 Voltage reference using the beta-multiplier.

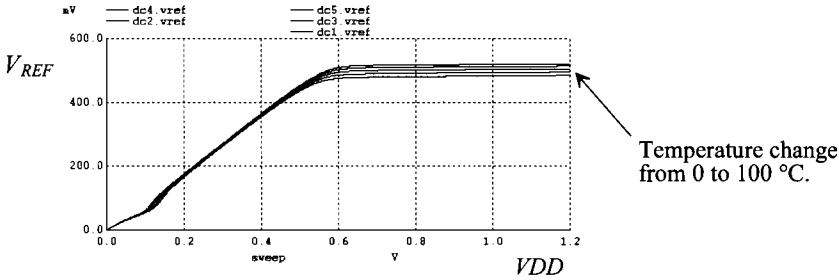


Figure 23.14 Simulating the operation of the reference in Fig. 23.13.

A couple of practical notes are needed at this point. While we've used simulations to zero in on a design with decent temperature behavior and a reference voltage at $V_{DD}/2$, actually fabricating this reference would result in a V_{REF} different from the one seen in Fig. 23.14 (in most fabrication runs). The MOSFET characteristics and the sheet resistance vary with each process run.

To adjust the reference voltage to $V_{DD}/2$, the resistor must be trimmed using fuses, Fig. 23.15. It can be shown that the temperature behavior doesn't vary significantly with small changes in V_{REF} when trimming. As seen in Eq. (20.38), lowering the resistor value causes V_{REF} to increase, while increasing the resistor value causes V_{REF} to decrease. The fuses in Fig. 23.15 short across the resistor until they are blown. To trim the resistor, we start blowing the resistors (electrically or with a laser). With each blown fuse we add (nominally) $200\ \Omega$ in series with the nominally 4k resistor. If the processes' sheet resistance increases by 20% (so that the resistors are now 4.8k and $240\ \Omega$), then we only need to blow three fuses (neglecting the change in the MOSFET characteristics) to trim the resistor to 5.5k . If the sheet resistance decreases by 20%, then the resistors are 3.2k and $160\ \Omega$. Fifteen fuses would need to be blown to trim the resistor to 5.5k .

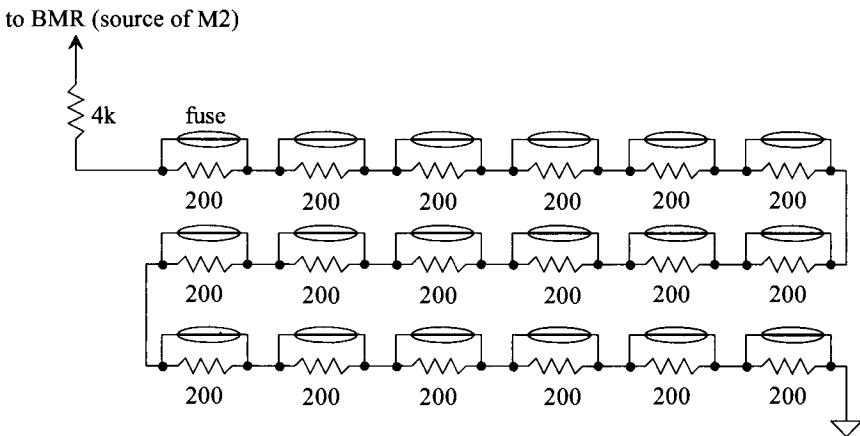


Figure 23.15 Trimming a resistor with fuses.